

The empirical recurrence for OEIS A298277, recovered from its tail

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Abstract

OEIS A298277 records “Empirical recurrence of order 92” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 142$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 97$. Its last coefficient is $c_{92} = 163857312 \neq 0$, so no further tail satisfies anything shorter; any order-92 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A298277 is “Number of $n \times 5$ 0..1 arrays with every element equal to 1, 2, 3, 4 or 5 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

3, 236, 6013, 136006, 3214459, 75989207, 1794078025, 42359577845, ...

The entry states:

Empirical recurrence of order 92 (see link above)

As of the “Last modified” line on the live entry (Jan 16 2018 08:06:51, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 142$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 142$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 92: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 92.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 284$ exact terms from $n = 5$ on, it returns order 92 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{92} c_i a(n-i),$$

c_1	21	c_{24}	−14905095618762	c_{47}	520010046385033	c_{70}	−171169846193271
c_2	88	c_{25}	−14397789175779	c_{48}	−5850168387044678	c_{71}	5881214363469
c_3	−417	c_{26}	37500493793961	c_{49}	−1650767719531087	c_{72}	55787752818717
c_4	−4552	c_{27}	82960637263291	c_{50}	3115015872202646	c_{73}	−17281158496034
c_5	−9995	c_{28}	−30848785348819	c_{51}	1028888026630608	c_{74}	6837881878425
c_6	61021	c_{29}	−254618863832724	c_{52}	816725201796807	c_{75}	27152781470245
c_7	305458	c_{30}	−173097515376122	c_{53}	1593092691290462	c_{76}	−11898357301869
c_8	622489	c_{31}	444534219002223	c_{54}	−2102266933130773	c_{77}	−12448926014443
c_9	−3089175	c_{32}	838290912979065	c_{55}	−2389996008030565	c_{78}	6705470060739
c_{10}	−17816792	c_{33}	−200396197864611	c_{56}	1712359653547285	c_{79}	1052948718148
c_{11}	−4760298	c_{34}	−1782531745545416	c_{57}	174725777826729	c_{80}	−2016592506570
c_{12}	159384311	c_{35}	−1136289425489331	c_{58}	−2827345849831976	c_{81}	1759058538533
c_{13}	368939850	c_{36}	1962172748005797	c_{59}	941839993146928	c_{82}	1021258818576
c_{14}	−843321534	c_{37}	3279130019197882	c_{60}	2991625777362128	c_{83}	−711055048267
c_{15}	−3930499936	c_{38}	−112472428233602	c_{61}	−868435376567655	c_{84}	−274342029711
c_{16}	1016120899	c_{39}	−4596566451190260	c_{62}	−1652034336274897	c_{85}	97428341836
c_{17}	35727516299	c_{40}	−3223807727275923	c_{63}	944659654035757	c_{86}	−8528355324
c_{18}	24067086433	c_{41}	3444497137583984	c_{64}	605206144900262	c_{87}	−9326424005
c_{19}	−215904954831	c_{42}	5476189427261829	c_{65}	−636559637169740	c_{88}	8507039993
c_{20}	−455646517773	c_{43}	−936881179783329	c_{66}	−174371653948609	c_{89}	−2902687792
c_{21}	848112056260	c_{44}	−5846758659101289	c_{67}	289107035536435	c_{90}	−1944045324
c_{22}	3356206085328	c_{45}	−137252789452092	c_{68}	106900378124162	c_{91}	438754648
c_{23}	−213172707460	c_{46}	6214196937169440	c_{69}	−131428479299595	c_{92}	163857312

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 97$, and it is the recurrence OEIS A298277 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{92} - c_1x^{92-1} - \dots - c_{92}$. Its constant term is $-c_{92} = -163857312$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 92 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 92, so $\deg q = 92$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 92, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 92, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 163857312.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-92 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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